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Circular water–energy integration in off-grid viticulture: An Ecocanvas approach to resource efficiency

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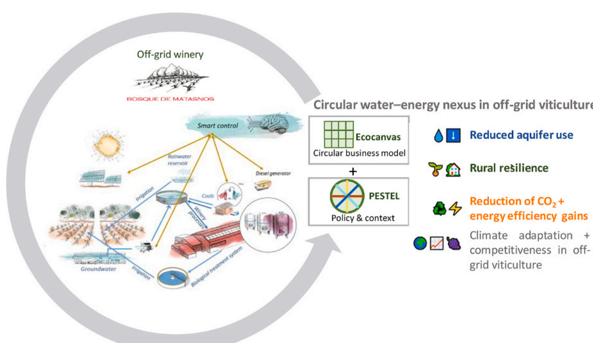
HIGHLIGHTS

- Circular water–energy integration tested in an off-grid Mediterranean winery
- Cascading water reuse reduces aquifer stress and enhances drought resilience.
- PV heat pumps saved 38 % heating and 56 % cooling energy, reducing diesel use by 80 %.
- Ecocanvas + PESTEL used as a systemic circular assessment framework
- Model connects climate adaptation with rural resilience and policy coherence.

GRAPHICAL ABSTRACT



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ABSTRACT

Viticulture faces growing challenges from climate change, including water scarcity, rising energy demand, and resource insecurity, particularly in off-grid rural areas. This study proposes a circular water–energy integration strategy to enhance resource self-sufficiency and resilience in agri-food systems. A mixed-methods framework was applied, combining technical monitoring of a circular water–energy system with business model analysis. The Ecocanvas framework, a circular adaptation of the Business Model Canvas, structured system integration, while a PESTEL analysis identified institutional and regulatory drivers. The system was evaluated under real operating conditions in an off-grid winery in Spain as part of the LIFE Climawin project, demonstrating both technical feasibility and circular performance. Results demonstrate technical feasibility and strong circular performance. Approximately 1000 m^3 of water were sequentially reused per production cycle, reducing groundwater extraction. Photovoltaic-powered water-to-water heat pumps improved thermal efficiency, achieving energy savings of 38 % for heating and 56 % for cooling compared with the baseline. Smart microgrid control enhanced overall system efficiency and stability, confining diesel use to backup operation only,

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corresponding to an inferred approximately 80 % decrease in diesel-generated electricity compared with the baseline, and lowering annual CO_2 emissions by about 260 t CO_2eq (−76.5 %). Overall, the study confirms that circular water–energy integration is a viable pathway to resource autonomy in off-grid viticulture. The findings highlight Ecocanvas as an effective decision-support tool for scaling circular transitions in small and medium-sized enterprises in the agri-food sector and for informing European Union sustainability policies.

1. Introduction

Viticulture is a cornerstone of Mediterranean agri-food systems, combining high economic value, cultural heritage, and territorial identity (Ruiz Pulpón and Cañizares Ruiz, 2022). Yet it also epitomizes the vulnerabilities of agriculture under climate change, particularly in semi-arid regions where vineyards face recurrent droughts, irregular rainfall, and rising energy demand. Reliable access to water is critical for vineyard irrigation and for maintaining stable thermal conditions during winemaking. Previous life-cycle studies have also highlighted the high water and carbon intensity of wine production, through analyses of water footprint and economic productivity (Miglietta et al., 2018), bottle-level carbon and water impacts (Bonamente et al., 2016), and full life-cycle assessments of Mediterranean wines (Fusi et al., 2014). It is also essential for hygienic and domestic functions within the winery and for sustaining broader ecosystem services. These interdependent uses underscore that water operates simultaneously as a production input and as a systemic resource, requiring management strategies closely coupled with energy flows to ensure long-term sustainability (Mirás-Avalos and Araujo, 2021).

Traditionally, sustainability in wine production has been framed around three pillars—social, economic, and environmental. However, recent studies emphasize the need to integrate energy as a fourth dimension, given its central role in production processes and its enabling function for circular transitions (Zambon et al., 2018). Fermentation cooling is one of the most energy-intensive stages of winemaking, although innovations such as optimized cooling strategies and tailored yeast strains have shown potential to reduce demand (Beghi et al., 2023). Moreover, water can also be conceived as a fifth pillar of sustainability in viticulture, since it is indispensable not only for vineyard irrigation but also for sanitation, cooling, cleaning, and maintaining broader ecosystem services. Its systemic role in viticulture goes beyond that of a mere production input, acting as a critical vector for resilience under climate stress. Therefore, climate-driven water stress and energy costs continue to reshape winery operations. Off-grid enterprises are particularly exposed due to their exclusive reliance on locally available resources (Sacchelli et al., 2017).

Research on water efficiency in viticulture has followed multiple lines. Deficit irrigation strategies have been tested to reduce water use while maintaining grape quality (Medrano et al., 2015), while reviews highlight the complexity of adapting vineyard water management to climate variability (Mirás-Avalos and Araujo, 2021). In Southern Europe, integrated approaches to water and wastewater management have been identified as essential for sustainability, combining irrigation efficiency with reuse strategies and wastewater valorization (Costa et al., 2020). Studies on wastewater reuse in vineyards demonstrate improved resilience to water scarcity (Drechsel et al., 2022), and rainwater harvesting has been proposed as a complementary resource (Costa et al., 2020). Recent work has also advanced design tools for optimizing storage capacity in such systems, supporting their wider applicability in agriculture (Fonseca et al., 2017).

Within this spectrum, water reuse has increasingly been explored as a strategy to mitigate scarcity, though its outcomes remain context-dependent. Tertiary treatment has been shown to support vine growth without compromising fruit quality (Petousi et al., 2019), whereas secondary effluents may introduce chemical and microbial hazards, requiring careful management of irrigation methods (Oron et al., 2001). Long-term trials in Israel—and more recent experiences in Spain—confirm that reclaimed water can provide essential nutrients and reduce pressure on primary sources (Ramírez and Ibáñez, 2023), but also pose risks of salinity buildup and sodium accumulation with potential effects on vine health (Simhayov et al., 2023). Broader reviews emphasize that these practices can close water gaps in agriculture if supported by adequate monitoring and treatment (Zhang and Shen, 2019). Complementary advances in natural greywater purification (Joshi et al., 2022) further resonate with Mediterranean applications, where decentralized treatment systems enhance the safe reuse of effluents and reinforce circularity in viticulture.

On the energy side, renewable technologies such as photovoltaic systems and heat pumps are increasingly adopted, alongside innovations in process-level efficiency (García et al., 2024). The use of groundwater as a thermal source in water-to-water heat pumps is well established in agro-industrial applications (Chen et al., 2023), with studies confirming improvements in energy efficiency under fluctuating conditions (Zhang et al., 2023). Recent analyses also highlight the influence of groundwater dynamics on thermal performance (Pandey and Basu, 2023) and provide analytical models to predict thermal plumes in groundwater heat pump systems (Pophillat et al., 2020).

Existing research on sustainability in viticulture has largely addressed water and energy as separate resource challenges, with studies focusing either on irrigation efficiency, wastewater reuse or renewable energy deployment. Although there is growing evidence of the benefits of water reuse and renewable integration, combined analyses of water–energy flows remain scarce, particularly under off-grid operating conditions where resource interdependence is most critical. Initial attempts to link water and energy efficiency have proposed joint indicators in agro-industrial systems (Carrasquer et al., 2017), and international work has highlighted the relevance of the water–energy nexus for policy and communication (Kenway et al., 2019). However, these approaches remain predominantly techno-centric and do not provide operational frameworks to translate nexus strategies into circular business models capable of enhancing resource autonomy in Mediterranean viticulture (Provenzano et al., 2024). In addition, smart energy management has received limited attention despite its relevance for coordinating renewable generation, battery storage and flexible loads to reduce diesel dependence in hybrid microgrids (Kumar, 2025). No previous studies have integrated smart control, water reuse and renewable energy within a circular system perspective or linked them to business model transformation.

Building on these insights into the water–energy nexus, circular economy approaches provide a broader framework to integrate resource efficiency with innovation and socio-environmental value creation in viticulture. In agri-food systems more generally, circularity is gaining traction as a strategy for sustainability (Sehnem et al., 2020). Yet in viticulture, efforts have largely focused on waste valorization—such as reusing grape marc or lees (Provenzano et al., 2024)—while the extension of circular principles to water–energy integration remains underdeveloped. Recent studies confirm that wine sector assessments still concentrate primarily on product-level environmental footprints (Frasnetti et al., 2024) and highlight methodological fragmentation across life-cycle approaches (Zambelli et al., 2023), underscoring the need for more systemic frameworks that integrate water and energy flows within circular business models.

Within this context, the Ecocanvas methodology has emerged as a valuable tool for operationalising circular principles in business models. Unlike the traditional Business Model Canvas (Osterwalder and Pigneur, 2010), Ecocanvas incorporates regenerative resource flows, socio-environmental value creation and resilience principles (Daou et al., 2020). Its visual and adaptable structure makes it suitable for small and medium-sized enterprises in resource-constrained settings, such as wineries, where business model innovation must integrate local ecological constraints. Recent scholarship highlights the relevance of systemic indicators to support circular transition (Das et al., 2022) and warns against rebound effects or superficial circularity (Velenturf and Purnell, 2021), which reinforces the need for methodological frameworks capable of linking technical performance with strategic decisions. To address this gap, this study combines Ecocanvas with a PESTEL (Political, Economic, Social, Technological, Environmental, Legal) analysis, which examines political, economic, social, technological, environmental and legal drivers that shape the feasibility and scalability of circular practices. Integrating both tools provides a multilevel analytical approach that connects business model design with socio-institutional context—an aspect frequently overlooked in circular economy applications in agriculture.

This paper examines the case of Bosque de Matasnos, an off-grid winery in Spain's Ribera del Duero region that has developed a holistic strategy combining water self-sufficiency, renewable energy autonomy, and biodiversity stewardship. The winery has adopted a triple water system and a photovoltaic-based microgrid to minimize reliance on diesel and to operationalize the water–energy nexus through context-sensitive circular practices.

The aim of this study is twofold:

(1) to demonstrate the operational feasibility of a circular water–energy system implemented in an off-grid winery under real conditions; and
(2) to assess its systemic viability by analysing how technical performance interacts with business model configuration and institutional drivers using a combined Ecocanvas–PESTEL framework, in order to evaluate its applicability and replicability in agri-food contexts.

This study addresses three gaps identified in recent international literature. First, most research in viticulture and circular economy has focused on single-resource strategies—such as waste valorisation or isolated water or energy efficiency measures—while integrated circular water–energy systems remain largely unexplored, especially under off-grid operating conditions. Second, although the water–energy nexus has gained increasing attention, existing studies tend to adopt predominantly technical perspectives, with limited integration of business model configuration and value creation strategies, which are essential for circular implementation. Third, institutional feasibility and scalability are still weakly assessed in the literature, as regulatory, socio-economic and territorial drivers are often treated as external constraints rather than as components of system design.

Accordingly, this study makes three original contributions.

(1) It provides empirical evidence of the operational feasibility of a circular water–energy system implemented under real off-grid viticulture conditions, a configuration rarely examined in previous studies.
(2) It introduces a combined Ecocanvas–PESTEL analytical framework, offering a multilevel method that connects resource integration with business model strategy and institutional context—overcoming techno-centric approaches prevalent in the field.
(3) It demonstrates systemic circularity in practice, showing that water–energy integration can function as a replicable circular business model that strengthens resource autonomy, climate resilience and rural sustainability.

The remainder of the article is structured as follows. Section 2 presents the case study and outlines the methodology, highlighting the integration of Ecocanvas and PESTEL. Section 3 reports the main results derived from technical monitoring and participatory analysis. Section 4 provides a critical discussion of the findings in relation to the literature and sectoral challenges. Finally, Section 5 synthesizes conclusions and policy implications.

2. Methodology

2.1. Case study: Bosque de Matasnos winery

Bosque de Matasnos is located in a semi-natural area near Peñaranda de Duero (Burgos, Spain), on the northern edge of the Ribera del Duero Denomination of Origin. The estate comprises 74 ha of vineyard, 80 ha of forest, and 50 ha of cereal, legume and oilseed crops, which are used in rotation to improve soil fertility and structure through organic matter recovery and biological nitrogen fixation, complemented by livestock and beekeeping activities (see Fig. 1 for location and land distribution). Since 2019, the winery has pursued a transition toward diversified and sustainable land management, replacing rainfed cereals with vineyards, expanding forest cover, and integrating agro-ecological practices. Legume rotations play a particularly strategic role by fixing atmospheric nitrogen, thereby enhancing soil fertility and reducing dependence on external fertilizers.

A guiding principle of Bosque de Matasnos is the self-sufficiency of water and energy, aiming to manage 100 % of resources on-site, minimize consumption, and return water to the soil. The estate has progressively adopted a circular approach, in which irrigation no longer depends solely on groundwater but is increasingly supplied through rainwater harvesting and the reuse of treated effluents, thereby reducing pressure on aquifers and reinforcing long-term resilience. The same philosophy applies to energy: the estate remains fully off-grid and has evolved from a fossil fuel–dependent operation into a hybrid stand-alone system based on photovoltaic generation with battery storage. Diesel units are now retained only as an emergency backup, while intelligent control ensures the optimal use of renewable resources. To reinforce transparency in resource management, Bosque de Matasnos has monitored its carbon footprint since 2013 and, more recently, began calculating its water footprint in 2024, which is currently undergoing external certification.

Climatic conditions in the Ribera del Duero—cold springs, autumns, and winters contrasted with hot, dry summers—further highlight the need for this integrated approach. Rising temperatures have increased both water stress and energy demand, particularly in summer, when the vineyard enters the phase of fruit growth and ripening at a time when rainfall is minimal, temperatures are highest, and irrigation requirements intensify. At the same time, extending into early autumn, irrigation needs coincide with cooling demands in the winery during harvest and fermentation. This overlap of water and energy peaks makes Bosque de Matasnos a strategic site for testing circular water–energy solutions under real operating conditions.

2.2. Circular water–energy model

The circular water–energy model at Bosque de Matasnos translates the winery's principle of resource self-sufficiency into a set of infrastructures and processes that maximize efficiency and reuse under off-grid conditions (see Fig. 2). The system integrates three complementary water sources into a closed loop:

✓ Rainwater reservoir (4000 m^3): stores blue water captured from rainfall, primarily for vineyard irrigation. Before application, the water is filtered through sand to remove suspended particles (e.g., leaves, dust, and debris) and avoid clogging of valves.
✓ Groundwater reservoir (1000 m^3): extracts blue water from an aquifer, which is subjected to sequential treatment—including

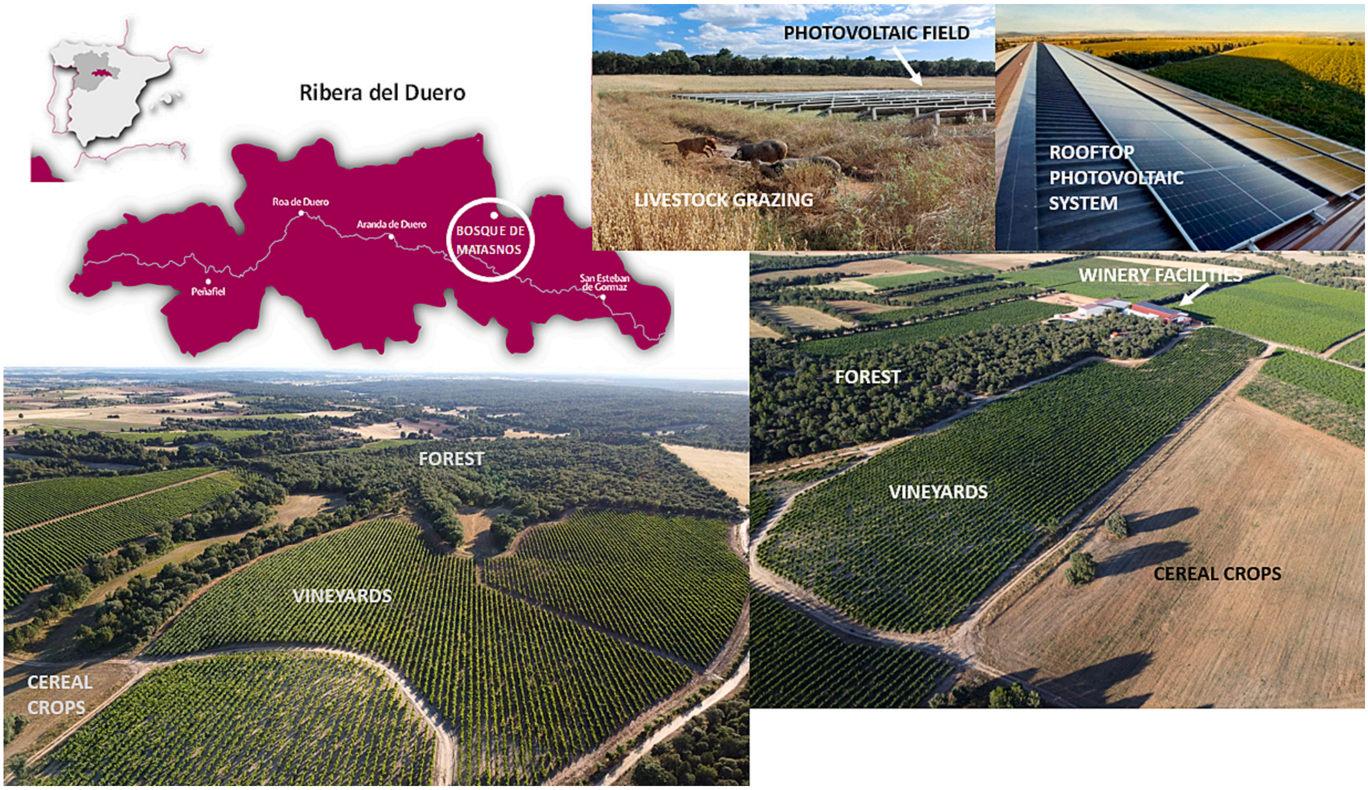

Fig. 1. Land and resource configuration at Bosque de Matasnos (Spain), combining vineyards, Mediterranean forest, crops, livestock and photovoltaic energy in an integrated circular system.

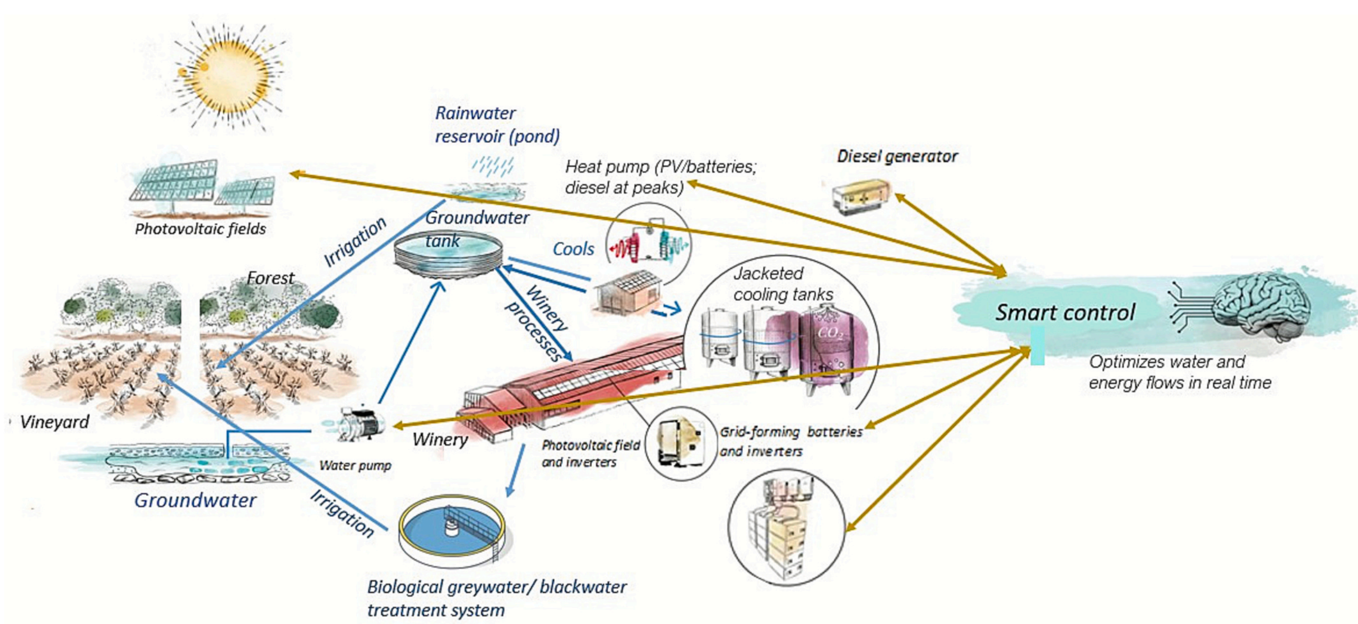

Fig. 2. Integrated circular water–energy model at Bosque de Matasnos winery.
The model integrates three complementary water flows. Rainwater is stored in a 4000 m^3 reservoir for vineyard irrigation. Groundwater is kept in a 1000 m^3 tank and used for winery processes such as cooling, cleaning, and sanitation, with part of the flow reused after thermal use. Greywater and blackwater are treated in a biological facility without chemical additives, allowing safe reuse for irrigation once authorized. These water flows are coupled with a photovoltaic microgrid, supported by batteries and emergency diesel generators, and managed by a smart control system that optimizes water and energy use in real time.

filtration, chlorination, decalcification, and final chlorination—until it reaches a quality suitable for winery operations (process cooling, cleaning, sanitation) and limited domestic uses. Once consumed, this water is converted into greywater and blackwater.

✓ Biological treatment facility: greywater from cleaning operations and blackwater from sanitary uses are treated on-site using an aerobic activated sludge process with prolonged aeration, followed by secondary sedimentation and sand filtration for effluent polishing

(Christou et al., 2024). This engineered biological treatment promotes aerobic microbial degradation of organic matter and achieves effluent quality suitable for controlled agricultural reuse, in line with European Union (EU) Regulation 2020/741 on minimum requirements for water reuse and Spanish Royal Decree 1620/2007. Reuse of treated effluent is subject to authorisation by the Confederación Hidrográfica del Duero (CHD), which oversees compliance and monitoring. The treated water is currently reused on pastures and will be extended to vineyard irrigation once formal approval is granted.

Together, these infrastructures ensure that water is reused sequentially across industrial, domestic, and agricultural functions, multiplying its utility before ultimately returning it to the soil. This circular loop is complemented by a seasonal wetland at the estate entrance, which remains active for around six months per year and provides habitat for birds, insects, and other species, reinforcing the biodiversity value of the estate.

Energy management is tightly integrated with this water loop through a stand-alone hybrid microgrid that supplies all electricity demands of both the winery and the vineyard. The system comprises two photovoltaic fields with a combined peak capacity of 248 kWp (179.8 kWp on the winery rooftop and 67.8 kWp in irrigation parcels), four grid inverters (200 kW), and a lithium-ion battery bank of 462 kWh supported by 21 inverter–chargers (≈105 kW). Two diesel generator sets (200 kVA, ≈160 kW) are retained exclusively as an emergency safeguard (see Table 1). The inverter–chargers and batteries form a stable three-phase grid that enables the coupling of the photovoltaic (PV) systems, while surplus generation is stored and later released during low-irradiance periods, reducing the need for diesel. Overall operation is coordinated by an advanced smart control system that prioritizes renewable generation, synchronizes it with water pumping, treatment, irrigation, and winery processes, and optimizes battery charging and discharging in real time. Before deployment of this control layer, diesel generators were frequently required to cover simultaneous demand peaks; today, intelligent management has reduced their use by around 80 %, significantly cutting fuel consumption and associated CO_2 emissions while ensuring a stable, resilient, and efficient energy supply (see Escriche et al., 2024, for more details).

2.3. Ecocanvas-based methodology

This research was structured around the Ecocanvas as the central methodological axis, adapting its application to the operational diagnosis and redesign of the circular water–energy model at Bosque de Matasnos. Beyond its theoretical role as a circular economy reference tool, the Ecocanvas was deployed as a practical framework to integrate quantitative measurements with qualitative insights.

The process unfolded in three iterative stages:

(i) *Mapping of resource flows* and system boundaries through field data (energy demand, water extraction, treatment processes, reuse volumes, and storage capacity) and technical documentation.
(ii) *Participatory reflection with winery managers* to identify constraints, enabling factors, and alignment with strategic priorities, particularly regarding the multifunctional use of water (cooling, sanitation, irrigation) and renewable-based autonomy.
(iii) *Redesign of business model canvas blocks* to explicitly incorporate cascading and sequential water use, renewable-powered thermal management, and indicators of circular performance and territorial value.

This multi-layered approach allowed the Ecocanvas to operate simultaneously as a diagnostic tool—quantifying efficiency gains, resource synergies, and circularity levels—and as a strategic instrument—structuring organizational learning, integrating stakeholder perspectives, and assessing long-term sustainability pathways. Its adaptability was critical to capture not only technical outputs (e.g., water reuse volumes, reductions in generator reliance) but also socio-economic implications and governance challenges inherent to operating under off-grid conditions.

2.4. Data collection and integration

The performance assessment of the Bosque de Matasnos circular system required analysing both the multifunctional use of water and its integration with renewable-based energy. Water flows were monitored across industrial, sanitary, domestic, and agricultural functions, including its role as a thermal vector for fermentation cooling—one of the most energy-intensive stages of winemaking.

To evaluate thermal management, different operational scenarios were defined for the heat pump system, with technical data consolidated from field monitoring, winery records, and system sensors. Indicators included Coefficient of Performance (COP), Energy Efficiency Ratio (EER), seasonal water availability, water reuse volumes, and the overall balance of the off-grid energy system. In parallel, the functioning of the microgrid was assessed, covering photovoltaic generation, battery storage, inverter–chargers, diesel backup units, and the smart control system. Key energy indicators included renewable fraction, diesel use, overall efficiency, and grid stability. These measurements were periodically reviewed in joint sessions with winery staff, technology partners, and the research team to ensure traceability, consistency, and reproducibility.

Complementary qualitative data were collected through semi-structured interviews and participatory workshops with the winery's owner, workers, and project partners. These interactions provided insights into organizational changes, enabling conditions, and perceived barriers to implementation, capturing aspects not measurable through technical indicators alone.

The integration of both datasets—quantitative and qualitative—ensured that the subsequent Ecocanvas analysis reflected not only technical performance metrics but also the broader contextual dynamics shaping the circular water–energy model at Bosque de Matasnos.

2.5. Methodological synthesis and contextual analysis

Based on the technical, organizational, and social information collected, the integrated framework was developed and represented graphically (see Fig. 3). This visual synthesis covers the twelve Ecocanvas blocks according to the original methodology.

The figure is introduced here as a methodological reference, while its detailed interpretation is provided in Section 3.3. A critical and in-depth

Table 1
Main components of the Bosque de Matasnos hybrid microgrid.

Component	Specification	Capacity	Notes
PV Modules – Winery rooftop	324 units	≈179.8 kWp	Mounted on coplanar structure (15°, E–W)
PV Modules – Irrigation parcels	189 units (45 + 144)	≈67.8 kWp	Fixed ground structure (18°, south-facing)
Grid inverters	4 units	200 kW	Located in winery facilities room
Inverter–chargers	21 units	126 kVA (≈105 kW)	Organized in 5 three-phase clusters
Battery system	30 units	462 kWh	Lithium-ion, low voltage, connected in parallel
Diesel gensets	2 units	200 kVA (≈160 kW)	Backup only, parallel connection
Smart control system	–	–	Synchronizes PV, batteries, gensets, and loads in real time

Source: Escriche et al. (2024).

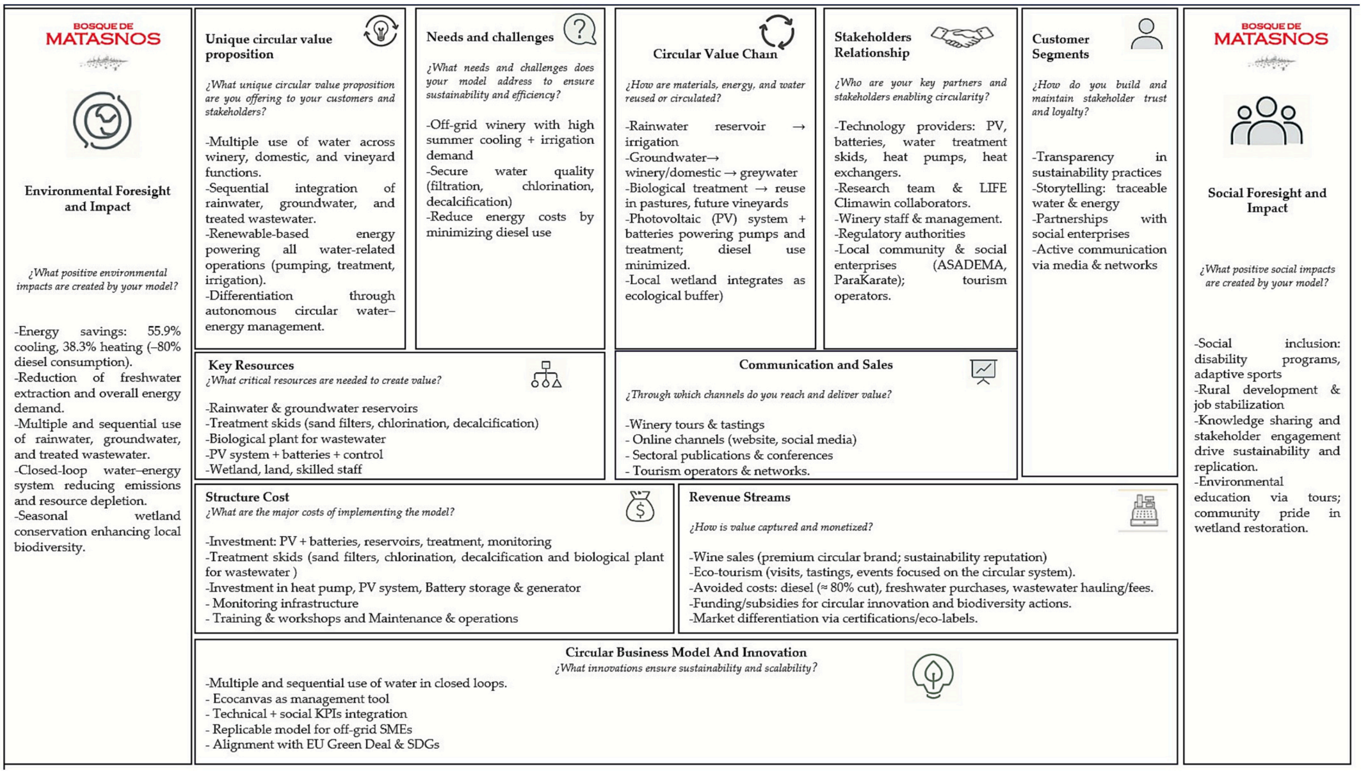

Fig. 3. Integrated methodological framework.
The framework combines quantitative system monitoring with qualitative inquiry, technical performance evaluation, and systemic assessment using Ecocanvas and PESTEL, leading to decision insights and policy implications for circular water–energy integration in off-grid viticulture.

analysis is provided for the most relevant blocks to characterize the impact, transferability, and replicability of the measure (value proposition, key resources, circular value chain, environmental and social impact, stakeholder relationships, cost structure, and innovation elements of the circular solution).

Contextual and strategic interpretation was further enhanced through the application of the PESTEL matrix. This approach enabled a rigorous, cross-cutting examination of the constraints and opportunities

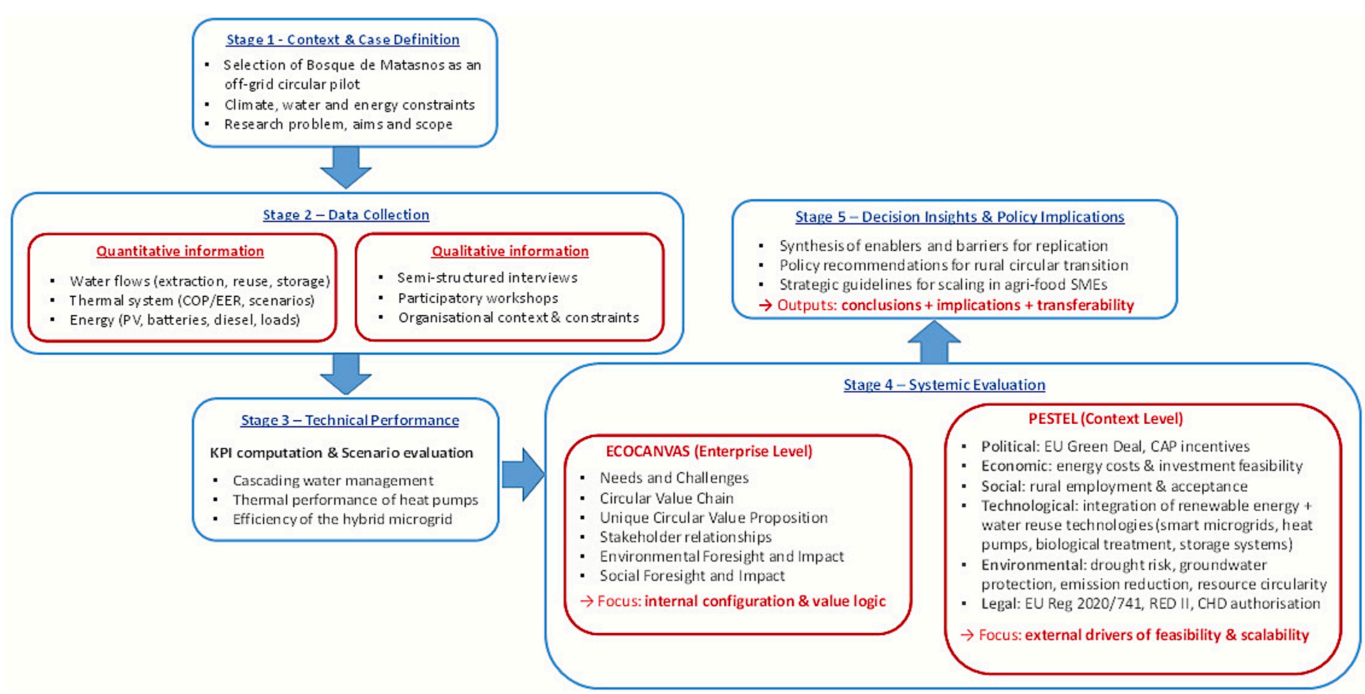

Fig. 4. Ecocanvas of the circular water–energy model implemented at Bosque de Matasnos.
The framework synthesizes the main blocks of value creation, resource flows, stakeholder relations, and impacts, highlighting the integration of water reuse and renewable-based energy under off-grid conditions.

affecting the sustainability and scalability of the measure in its sectoral and regulatory environment.

This integrated methodological framework (see Fig. 4) provides the analytical basis for the performance assessment in Section 3 and the subsequent discussion and interpretation of results in Section 4.

3. Results

The results are presented in four stages, following the methodological sequence. First, Section 3.1 examines the technical performance of the circular water–energy model, combining quantified efficiency indicators with evidence on resource flows, renewable integration, and water reuse. Next, Section 3.2 addresses organizational and social findings, highlighting how stakeholder engagement, partnerships, and community initiatives reinforced the viability and transferability of the model. These results complement the technical analysis by showing how circular strategies generate both social value and territorial resilience. Section 3.3 then offers a detailed interpretation of the Ecocanvas framework, emphasizing the blocks most relevant to circular value creation. Finally, Section 3.4 applies the PESTEL matrix to situate the findings within broader regulatory, institutional, and territorial contexts, identifying opportunities and constraints for replication and scalability.

3.1. Technical performance of the circular measure

The implementation of the circular water–energy system at Bosque de Matasnos winery forms part of the LIFE Climawin project, an EU-funded initiative that validates climate solutions under real operating conditions. Unlike laboratory testing, LIFE projects promote pilot-scale applications that demonstrate both technical feasibility and broader socio-environmental value. In this context, Bosque de Matasnos has become a pioneering case for circular water–energy integration in off-grid viticulture, showing how European policy instruments can accelerate the transition toward climate-smart agri-food systems.

From a technical perspective, performance assessment confirmed that the circular configuration ensures robust operation of both water and energy systems. On the water side, rainwater harvesting, groundwater extraction, and biological treatment were effectively combined in a cascading loop. Analytical results showed that the treated effluent meets quality standards for vineyard irrigation, although its application is currently restricted to pastures pending regulatory authorization. This configuration reduces dependence on aquifers, buffers seasonal variability, and enables each cubic meter of water to be reused multiple times before returning to the soil.

On the energy side, the hybrid microgrid demonstrated high levels of efficiency and resilience. The smart control system substantially reduced the reliance on diesel generators while maintaining stability and optimizing the use of renewable resources. Monitoring data also confirmed improvements in overall system efficiency, renewable fraction, and grid stability, with corresponding reductions in fuel consumption and CO_2 emissions.

The technical evaluation focused on three complementary dimensions: (i) thermal performance of heat pumps, (ii) efficiency of the hybrid microgrid, and (iii) cascading water management. Their key performance indicators are summarized in Table 2.

For thermal management, three operational scenarios were compared. Scenario 1 corresponded to a baseline of 100 % air-to-water heat pump, in which all thermal demand was met through aerothermal technology mainly applied in office and social areas. Scenario 2 was a hybrid configuration with 30 % air-to-water and 70 % water-to-water, where aerothermal equipment partially contributed but groundwater through a water-to-water heat pump covered most production processes. Scenario 3 was an optimized configuration with 10 % air-to-water and 90 % water-to-water. This last scenario was particularly effective during the three colder seasons (spring, autumn, and winter), when lower cooling demand allowed the water-to-water system to cover nearly the full load with only punctual aerothermal support. Scenario 3 proved the most efficient, delivering energy savings of 55.9 % for cooling and 38.3 % for heating relative to the baseline. Seasonal monitoring yielded an average EER of 5.38 and a COP of 3.95, confirming the technical viability of groundwater-based thermal management in off-grid conditions and its potential applicability in similar viticultural contexts (see Losantos and Valiño, 2025, for more details).

Table 2
Key performance indicators (KPIs) of the circular water–energy system at Bosque de Matasnos.

KPI	Value/scenario
Thermal performance (Scenario 3 (reference) – heat pump system)	
EER (water-to-water HP)	5.38
COP (water-to-water HP)	3.95
Energy saving – cooling (%)	55.9 % (vs. baseline)
Energy saving – heating (%)	38.3 % (vs. baseline)
Water reused per cycle (L)	1000,000
Drilling/additional infrastructure	Not required
Smart grid performance under Multi-Agent Hybrid Control (MAHC)	
Renewable fraction (%)	77.7 %
Overall efficiency (%)	59.2 %
CO_2 reduction vs. genset-only	≈260 tCO_2eq/year (≈76.5 %)
Additional CO_2 reduction vs. centralized control	≈12 tCO_2eq/year (≈14.3 %)
Diesel use	≈80 % (inferred reduction based on renewable fraction)
Grid stability (max voltage drop)	Δu ↓ 0.7 % (≈17.1 % reduction)
Water management	
Groundwater tank capacity	1000 m^3 (central source for winery operations)
Rainwater reservoir capacity	4000 m^3 (primary source for vineyard irrigation)
Reuse cycles per m^3	2–3 productive uses (cooling, cleaning/domestic, irrigation)
Water reuse volume	Several thousand m^3/year (effective saving vs. linear use)
Effluent treatment	100 % of greywater and blackwater processed in biological facility
Reuse application	Currently on pastures; validated for vineyards (pending authorization)
Discharge avoided	Wastewater effluent fully reused, no discharge to environment

Source: Escriche et al. (2024) and Losantos and Valiño (2025).

At the system level, the smart control scheme substantially improved the performance of the microgrid. Simulations showed that, compared to a genset-only baseline, renewable integration reduced CO_2 emissions by approximately 260 tCO_2eq per year (−76.5 %), while the introduction of the multi-agent smart controller added a further 12 tCO_2eq/year reduction (−14.3 %) relative to centralized control. Overall efficiency improved from 22.0 % in the genset-only baseline to 59.2 % with smart control. In parallel, the renewable fraction of the energy supply increased to 77.7 %. In addition, grid stability improved, with the maximum voltage drop decreased by 0.7 %—equivalent to a 17.1 % relative reduction—and diesel use was confined to backup operation only, corresponding to an inferred approximately 80 % decrease in diesel-generated electricity compared with the baseline (see Escriche et al., 2024, for more details).

Beyond energy efficiency, the system reinforces water circularity through a cascading use of groundwater, rainwater, and treated effluents. Each cubic meter of water is valorized two to three times across winery operations, domestic uses, and irrigation before returning to the soil, reducing aquifer withdrawals and avoiding wastewater discharge. Analytical results confirm that reclaimed water meets the standards for vineyard irrigation, although official authorization is still pending. While precise figures will be consolidated through the ongoing water footprint assessment, current evidence indicates substantial savings and increased resilience to drought conditions.

Nevertheless, certain limitations persist. Seasonal irregularities in rainfall reduce the contribution of harvested rainwater, and reliance on diesel generators during peak demand highlights the need for greater renewable integration. While three seasons (spring, autumn, and winter) present lower water pressure, summer is hot and very dry, with simultaneous peaks in irrigation requirements and cooling demand. In this context, flexible water management and resilient energy systems are essential to ensure long-term sustainability.

Overall, the Bosque de Matasnos system demonstrates that a circular configuration linking water reuse with renewable-based energy management can deliver robust technical performance, as reflected in the Key Performance Indicators (KPIs) presented in Table 2, while reducing environmental pressures. However, as the following section illustrates, the sustainability of such measures depends not only on technical indicators but also on their alignment with organizational strategies, stakeholder engagement, and territorial dynamics.

3.2. Additional findings: social and organizational aspects

Beyond technical performance, the participatory process revealed a set of organizational and social dimensions that are central to the sustainability and transferability of the circular water–energy model. These findings, derived from semi-structured interviews, workshops, and document analysis, complement the quantitative assessment and directly inform the stakeholder and social impact blocks of the Ecocanvas.

A first dimension concerns social inclusion and partnerships. The winery maintains long-term collaborations with organizations dedicated to social integration, channeling part of its revenues into initiatives that support people with disabilities and adaptive sports. Although not captured in standard employment metrics, these efforts represent a broader form of value creation that extends beyond the winery itself, aligning with an expanded understanding of circular economy practices where value also includes social equity and community resilience. In this sense, they reinforce recent literature highlighting the importance of articulating social value in agri-food circular models (Atanasovska et al., 2022) and respond to calls for a more systematic integration of equity into circular business strategies (Valencia et al., 2023).

At the territorial level, the winery positions its activities as contributors to rural development and local revitalization. The municipality where it is located is severely affected by depopulation, with limited services and economic opportunities. In this context, employment in the winery and vineyards, partnerships with local suppliers, and the promotion of wine tourism play a strategic role. Guided visits, training sessions, and corporate events organized at the estate generate additional demand for the few local accommodations, shops, and bars that remain in the area, thereby supporting the survival of small businesses. These actions reinforce the role of agri-food enterprises as anchors of territorial resilience, consistent with findings that highlight stakeholder involvement as a driver of circular and sustainable agri-food supply chains (Andika et al., 2025).

Knowledge transfer and awareness raising also emerged as relevant aspects. Through technical workshops, educational visits, public events linked to LIFE Climawin, and collaborations with universities and business schools, the winery disseminates lessons learned and fosters transparency. Permanent informational panels in the vineyards and forest ensure that visitors can access details about circular practices independently. These outreach efforts amplify visibility and generate systemic knowledge spillovers at both sectoral and territorial levels, consistent with evidence that knowledge management and open innovation are key drivers of competitiveness in the wine industry (Martínez-Falcó et al., 2024).

Finally, organizational culture and reputation were highlighted as intangible assets. Active communication through professional networks, specialized media, and social platforms strengthens the winery's image as a sustainability-oriented enterprise. This, in turn, builds trust among environmentally conscious stakeholders and positions Bosque de Matasnos as a reference point for integrated circular practices in the wine sector, consistent with recent findings that emphasize the role of green intellectual capital as a source of competitive advantage in Spanish wineries (Martínez-Falcó et al., 2025).

From a systemic perspective, the Bosque de Matasnos circular model cannot be assessed solely in terms of efficiency gains. Its success is equally tied to the integration of social practices, community engagement, and organizational culture, which collectively enhance resilience and create shared value. Positioning these dimensions within the Ecocanvas highlights their systemic relevance and underscores the importance of embedding social sustainability into circular economy transitions (Valencia, 2023).

3.3. Ecocanvas synthesis and analysis

Synthesizing technical, organizational, and social findings provides a systemic perspective on circular innovation (Suchek et al., 2021). The Ecocanvas framework was applied to integrate and structure the evidence collected, offering a comprehensive view of how circularity is operationalized at Bosque de Matasnos. Building on previous applications in the agri-food sector (Daou et al., 2020), Fig. 3 illustrates the Ecocanvas blocks, connecting the technical KPIs, social initiatives, and collaborative processes that underpin the winery's circular water–energy model.

The analysis highlights several blocks as particularly relevant. The value proposition combines the production of premium wines with a strong sustainability orientation, linking water–energy self-sufficiency to biodiversity stewardship and territorial revitalization. Key resources include not only infrastructures such as reservoirs, photovoltaic installations, and treatment facilities, but also technical know-how and long-term alliances. The circular value chain reflects the integration of cascading water use with renewable-based energy flows, demonstrating the operationalization of the water–energy nexus in an off-grid context.

Beyond technical aspects, the Ecocanvas captures the importance of stakeholder relationships, particularly alliances with technology providers, universities, and social organizations, which sustain innovation and enable knowledge transfer. Environmental and social impacts are also embedded, encompassing reduced emissions, biodiversity preservation, and inclusive initiatives that channel business revenues into local programmes. In this sense, the Ecocanvas makes explicit how technical innovation is coupled with community and territorial value creation (Valencia, 2023).

Complementing this synthesis, the application of the PESTEL matrix situates the model in its external context (Rahman and Mishra, 2023). Regulatory uncertainties regarding wastewater reuse, opportunities arising from European sustainability agendas, and socio-economic challenges linked to rural depopulation emerge as critical factors influencing scalability and replicability.

Overall, the Ecocanvas synthesis demonstrates that Bosque de Matasnos is not simply a technical pilot, but an integrated circular strategy. Its contribution lies in showing how multifunctional water reuse and renewable-based energy can be embedded within a coherent business model that aligns efficiency gains with ecological stewardship and social inclusion. Beyond its practical value, the case advances academic debate by illustrating how the Ecocanvas can capture the integration of technical, social, and ecological dimensions into systemic circular economy frameworks.

3.4. Detailed block analysis

To unpack the operational logic of the circular solution, the following subsections examine the Ecocanvas blocks in depth. The analysis begins with the structural needs and constraints of the winery, before moving through value creation processes, stakeholder networks, and external drivers, highlighting how each element interacts to form a coherent, climate-adaptive system.

3.4.1. Needs and challenges

As discussed in Sections 2.1 and 3.1, Bosque de Matasnos is a medium-scale, off-grid winery operating in a semi-arid context, which creates specific pressures for water and energy management. Identifying these structural needs and challenges explains why the adoption of a circular model was required.

First, the absence of connection to national electricity and sewer networks forces the estate to secure autonomy in both energy and water services (Kaundinya et al., 2009). Although renewable integration and smart control have significantly reduced diesel dependence (see KPIs in Table 2), fossil backup remains necessary under peak conditions, with implications for emissions and cost exposure (Katalenich and Jacobson, 2023).
Second, seasonal overlaps in resource demand—particularly in summer, when irrigation coincides with fermentation cooling—intensify stress on both water and energy systems. This interdependence highlights the need for synchronized management of flows rather than isolated solutions.
Third, climate variability in Mediterranean viticulture has exacerbated aquifer stress (Sacchelli et al., 2017). While rainwater harvesting and wastewater treatment have been introduced, regulatory restrictions still limit effluent reuse in vineyards (Provenzano et al., 2024), delaying the full closure of the cycle.
Finally, broader economic and market dynamics amplify these challenges. High upfront costs restrict access to advanced technologies such as deep geothermal (Thorsteinsson and Tester, 2010), while premium wine consumers demand transparency in water and energy footprints (Aivazidou and Tsolakis, 2020). At the same time, European policies, including the Green Deal and Circular Economy Strategy, increasingly push agri-food businesses toward verifiable circular practices (Valencia et al., 2023).

In short, these needs and challenges justify the systemic circular approach adopted at Bosque de Matasnos. More than a technical adaptation, it constitutes a strategic response that combines operational efficiency with environmental stewardship, territorial resilience, and consumer trust.

3.4.2. Circular value chain

The circular value chain analyzed in this study is based on the integrated water–energy model implemented at Bosque de Matasnos within the LIFE Climawin project. Examined through the Ecocanvas, it illustrates how resource flows are connected rather than managed separately. This approach represents a concrete technical innovation: the joint management of water and energy. In doing so, it delivers environmental, operational, and socio-economic value for viticulture systems (Marques et al., 2025).

As reported in Section 3.1, water is secured from three complementary sources—groundwater, rainwater, and treated effluents—-which are sequentially allocated to industrial, domestic, and agricultural functions before ultimately returning to the soil. This cascading use reduces aquifer withdrawals and strengthens resilience to climatic variability (Provenzano et al., 2024).

Energy is embedded in this cycle through an off-grid photovoltaic microgrid with battery storage, complemented by diesel units operating only as emergency backup. A smart control system synchronizes renewable generation with pumping, treatment, and cooling needs, ensuring stability during seasonal peaks such as summer irrigation and

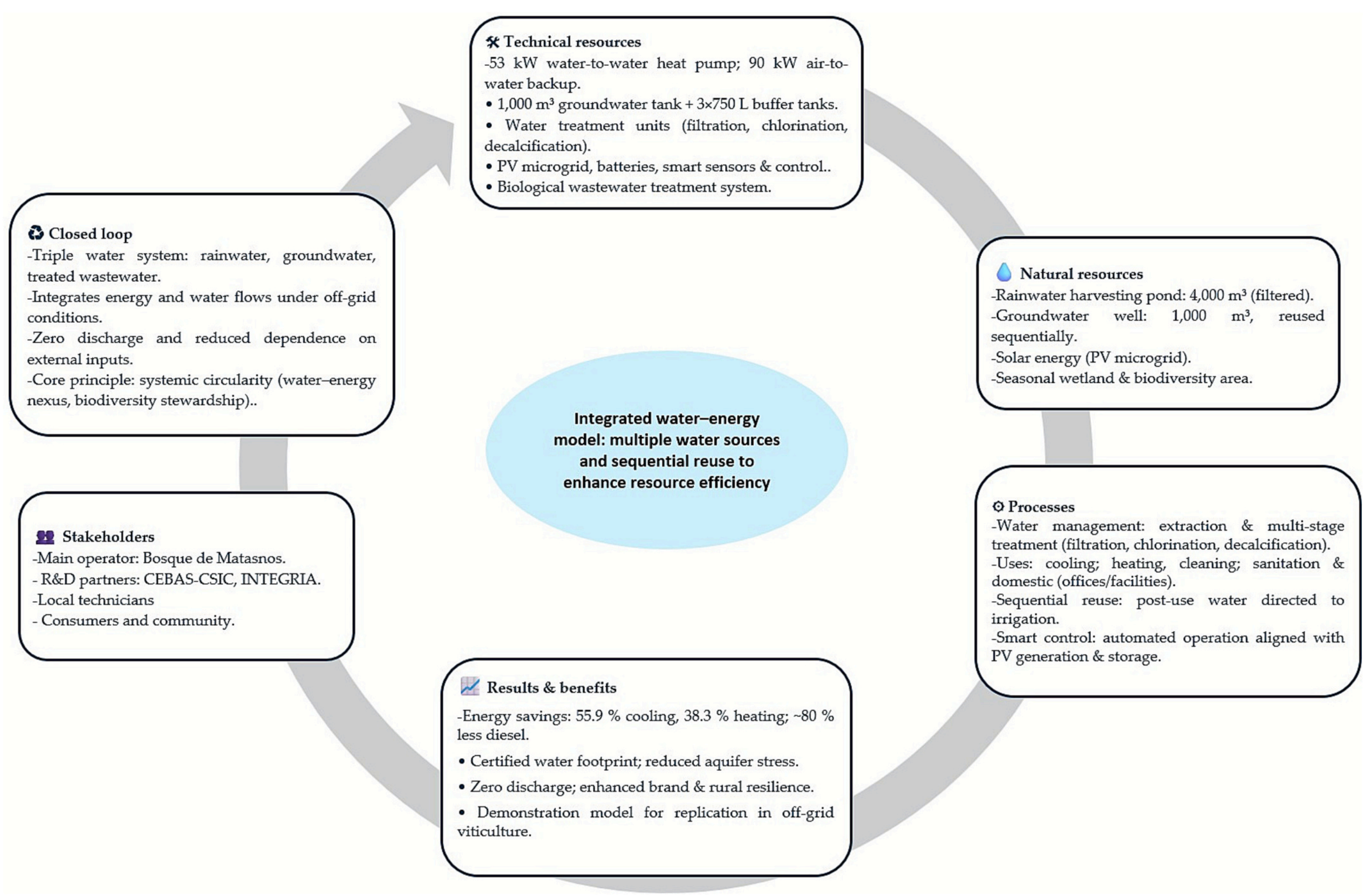

Fig. 5. Circular flow diagram (Ecocanvas) for reusing irrigation water as a thermal source in an off-grid winery.

fermentation (Beghi et al., 2023).

By linking water reuse with renewable energy in a coherent cycle, the Bosque de Matasnos model exemplifies how a circular value chain can operationalize the water–energy nexus under off-grid conditions. Beyond efficiency, the model contributes to market differentiation, stakeholder engagement, and sectoral learning, as further detailed in Section 3.2 (Poponi et al., 2023).

The main elements and resource flows of this circular value chain are summarized in Fig. 5, which illustrates the triple water strategy, renewable energy integration, and cascading use of resources that link technical efficiency with environmental stewardship and stakeholder value creation. These insights are explicitly captured in the Ecocanvas blocks, reinforcing not only the systemic relevance of the Bosque de Matasnos model but also its potential scalability and replicability within circular agri-food systems.

3.4.3. *Unique circular value proposition*

The unique circular value proposition developed at Bosque de Matasnos emerges from the integration of an off-grid water–energy reuse system that simultaneously enhances environmental performance, operational resilience, and socio-economic value creation. Unlike the descriptive mapping of flows in Section 3.4.2, this subsection emphasizes why the configuration provides a distinctive competitive and sustainability advantage—linking technical efficiency with broader societal and sectoral benefits.

From the perspective of consumers and markets, the winery delivers premium wines produced under demonstrably reduced water and energy footprints. Monitoring and validation within the LIFE Climawin project provide robust evidence of performance, strengthening transparency and meeting international demand for traceability and verifiable sustainability standards in agri-food chains (Boix-Fayos and De Vente, 2023). These outcomes are grounded in the quantified indicators presented in Table 2, which integrate both water and energy performance — including cascading reuse, reductions in aquifer withdrawals, efficiency gains in cooling and heating, a higher renewable fraction, and significant cuts in CO_2 emissions.

For the local community, the model contributes to aquifer conservation and resilience against climatic variability while generating socio-economic co-benefits through employment, wine tourism, and educational outreach. These dimensions illustrate how circular resource strategies can reinforce rural development while safeguarding environmental assets (Boyd et al., 2022).

For supply chain partners, the system offers a technically validated configuration that achieves major reductions in resource use. In energy terms, it cuts consumption by more than 50 % in cooling and one-third in heating compared with baselines, without requiring additional drilling or costly infrastructure. In water terms, it enables cascading reuse across winery operations, domestic functions, and irrigation, thereby reducing withdrawals from the aquifer and avoiding discharge. Taken together, these outcomes lower technological and financial barriers to adoption while aligning suppliers and technology providers with emerging Environmental, Social, and Governance (ESG) criteria and procurement frameworks demanded by global markets (Gurung et al., 2025). The robustness of this configuration is further supported by the performance indicators in Table 2, which integrate water and energy metrics to demonstrate both technical efficiency and circular resource use.

For policy and sectoral stakeholders, Bosque de Matasnos demonstrates how circular business models can reconcile climate objectives with competitiveness under real off-grid conditions. Within LIFE Climawin, the winery pilots a Smart Renewable Energy approach that integrates photovoltaic generation, battery storage, and controlled loads across vineyard, winery, farming, and enotourism facilities.

Extrapolating from the monitored efficiency improvements of the water-to-water heat pump configuration (55.9 % savings in cooling and 38.3 % in heating, see Table 2), the system achieves an estimated ≈200 MWh of annual energy savings. In parallel, around 1000 m^3 of water are reused per production cycle through cascading reuse across winery operations, domestic functions, and irrigation. If extended to the ~300 wineries of the Ribera del Duero Denomination of Origin, such measures could generate reductions of ≈60 GWh of energy and ≈300 million liters of freshwater annually. While indicative and subject to heterogeneity, these figures illustrate the potential cumulative contribution of circular water–energy systems to European Green Deal objectives (Boix-Fayos and De Vente, 2023). It should be noted that the detailed water management indicators in Table 2—such as reservoir capacities and reuse cycles—are specific to Bosque de Matasnos and cannot be linearly scaled to other wineries, given infrastructural and operational differences across the sector.

The Bosque de Matasnos case therefore demonstrates how circular strategies can create multidimensional value: improving efficiency, strengthening brand positioning through verified sustainability claims, supporting community development, and offering a replicable benchmark for policy and sectoral learning. These outcomes are directly reflected in the value proposition block of the Ecocanvas, linking competitive advantage with systemic transformation.

3.4.4. *Stakeholder relationships*

Within the LIFE Climawin framework, stakeholder relationships have been decisive for the design, implementation, and validation of the Bosque de Matasnos circular model. Engagement is not treated as a peripheral communication task but as a structural component of the innovation process, ensuring transparency, scientific rigor, and operational robustness (Kujala et al., 2023).

The governance structure integrates organizations with complementary expertise. Bosque de Matasnos leads operational deployment and demonstration under real production conditions. The Instituto de Carboquímica of the Consejo Superior de Investigaciones Científicas (ICB-CSIC) coordinates the technical aspects, designs the geothermal heat exchanger system, and conducts scientific validation. The University of Zaragoza evaluates social and economic dimensions through surveys, workshops, and stakeholder consultations, while the Spanish Wine Technology Platform ensures dissemination and transfer to other wineries and value-chain actors (Hubeau et al., 2017).

These formal collaborations are complemented by engagement with regulatory authorities, certification bodies, and local suppliers. This ensures compliance with sustainability regulations, alignment with European policy frameworks, and the strengthening of regional socio-economic networks. The stakeholder block of the Ecocanvas thus reflects how collaborative governance and cross-scale partnerships underpin the systemic relevance of the solution.

Rather than a stand-alone innovation, Bosque de Matasnos operates as a living laboratory for sustainable viticulture, offering transferable lessons for other agri-food contexts. This perspective aligns with emerging frameworks on circular business model innovation and systemic transitions toward agri-food circularity (Geissdoerfer et al., 2020).

3.4.5. *Environmental foresight and impact*

While earlier subsections emphasized immediate efficiency gains, this section highlights the long-term environmental foresight embedded in the Bosque de Matasnos circular strategy. The system anticipates future constraints associated with climate change, water scarcity, and increasingly stringent agricultural regulations.

The first dimension concerns water security. In a semi-arid context where groundwater reserves face increasing pressure (Sacchelli et al., 2017), the winery has adopted a triple water strategy that combines rainwater harvesting, groundwater use, and biological treatment of effluents. As described in Section 3.1, this cascading configuration allows the same cubic meter of water to be used sequentially for cooling, cleaning and irrigation before returning to the soil through controlled infiltration. This approach reduces net groundwater extraction and enhances local water autonomy under drought conditions while remaining consistent with the principles of sustainable water management promoted by European Union (EU) Regulation 2020/741 on water reuse and the EU Water Framework Directive (Netti et al., 2024).

The second dimension relates to energy and carbon foresight. The winery operates through an off-grid photovoltaic microgrid with battery storage, complemented by diesel generators used only in emergency conditions. A smart control system coordinates renewable generation with the water-to-water heat pump, pumping, and cooling loads, reducing fossil fuel dependency and ensuring operational stability during seasonal peaks. This configuration anticipates stricter emission limits under the European Green Deal, while also enhancing autonomy, lowering CO_2 emissions, and mitigating exposure to energy price volatility—factors that will be decisive for long-term competitiveness (Prada et al., 2024).

Beyond water and energy, the system provides ecological co-benefits. A seasonal wetland at the estate entrance supports biodiversity, while crop rotations with legumes improve soil fertility and reduce external fertilizer inputs. Together with forest conservation, these practices buffer the estate against extreme climatic events, mitigate aquifer stress, and strengthen ecosystem services. Intelligent monitoring further ensures that resources are allocated precisely where and when required, maximizing ecological performance while minimizing waste.

These outcomes, reflected in the environmental and social impact blocks of the Ecocanvas, illustrate how circular water–energy systems can evolve from technical solutions into broader adaptation pathways. By linking resource efficiency with biodiversity stewardship and climate resilience, the Bosque de Matasnos case demonstrates how circular models anticipate future pressures while creating systemic value for both the environment and society.

This foresight perspective highlights that Bosque de Matasnos is not only achieving immediate resource efficiency, but is proactively adapting to future environmental and regulatory pressures. In doing so, the winery exemplifies how circular business models can function as long-term adaptation pathways that align operational practices with climate resilience and European sustainability targets.

3.4.6. *Social foresight and impact*

The social foresight dimension underscores how circular innovation in viticulture can act as a catalyst for rural resilience and demographic stabilization. In regions affected by persistent depopulation, resource self-sufficiency strategies generate employment opportunities that are inherently local—vineyard management, winery operations, wine tourism, and hospitality—and therefore cannot be relocated or outsourced (Martínez-Carrasco Pleite and Colino Sueiras, 2024).

At Bosque de Matasnos, this effect is reinforced through long-term collaborations with social enterprises and inclusion programmes, which channel part of the winery's revenues into initiatives supporting people with disabilities and adaptive sports. These partnerships not only generate social value but also build community cohesion, embedding circular practices within a broader territorial development strategy.

Knowledge transfer and capacity building provide another anticipatory dimension. Training programmes on sustainable viticulture, guided visits, and dissemination through the LIFE Climawin platform foster a culture of innovation and transparency. Permanent information panels in the vineyards and forest ensure that knowledge is accessible even without guided tours, while collaborations with universities, business schools, and gastronomic centers strengthen intergenerational learning and professional skills. Together, these initiatives contribute to a knowledge-based rural economy and position the winery as both a production site and a hub for innovation and awareness raising.

This territorial role aligns with evidence that viticulture, when embedded in sustainable and socially inclusive systems, can sustain landscapes and reinforce regional identity (Giordano, 2017). It also reflects broader debates on how circular business models integrate social innovation and equity into systemic transitions (Valencia, 2023). By anticipating demographic risks and embedding social foresight into its circular strategy, Bosque de Matasnos exemplifies how sustainability-oriented viticulture can become a lever for rural regeneration and demographic resilience in Europe's depopulating regions.

3.4.7. *PESTEL analysis*

To complement the technical, environmental, and social findings, a PESTEL framework was applied to situate the enabling and constraining conditions shaping the Bosque de Matasnos circular water–energy model (Fig. 6). The PESTEL complements the Ecocanvas by contextualizing internal dynamics within external systemic drivers. This analysis clarifies how political, economic, social, technological, environmental, and legal factors influence adoption and replicability in rural, off-grid viticulture.

Politically, the winery benefits from a favorable policy environment shaped by the European Green Deal and the Common Agricultural Policy (CAP) (2023–2027), both of which emphasize renewable energy, water circularity, and rural development (Singh, 2025). Groundwater extraction quotas enforced by the Duero Hydrographic Confederation also incentivize water reuse, while access to LIFE funding facilitated system implementation.

Economically, the Ribera del Duero's export orientation creates competitive pressure to meet sustainability standards. Rising energy costs and off-grid constraints intensify this challenge. For Bosque de Matasnos, LIFE Climawin resources helped offset high upfront costs, while diversification into premium wines and enotourism improved financial resilience (Perramon et al., 2024).

Socially, viticulture provides diverse local jobs and stimulates complementary activities such as tourism and education, aligning with strategies that view rural entrepreneurship as a response to depopulation (Jorge-Vázquez et al., 2024).

Technologically, the model combines water-to-water heat pumps, photovoltaic energy and smart energy control in a modular configuration. This facilitates adaptation to different production scales and demonstrates its viability as a replicable innovation for agri-food small and medium-sized enterprises (SMEs).

Environmentally and legally, the multi-source water system—combining rainwater harvesting, groundwater extraction and biological treatment for cascading reuse—reduces pressure on local aquifers and reinforces climate adaptation. The system adheres to the circular water principles established by European Union (EU) Regulation 2020/741 on water reuse and Spanish Royal Decree 1620/2007, which govern the quality and safe application of reclaimed water in agriculture. However, the deployment of reuse practices remains conditioned by lengthy authorisation procedures from River Basin Authorities (Confederación Hidrográfica del Duero), indicating a regulatory–administrative gap that delays circular innovation despite compliance with environmental standards. This legal barrier reflects broader challenges already noted at European level, where administrative delays and fragmented implementation frameworks continue to constrain large-scale implementation of water reuse (Christou et al., 2024). On the energy side, the model aligns with European decarbonisation objectives under the Renewable Energy Directive (RED II, Directive (EU) 2018/2001) and Spain's National Integrated Energy and Climate Plan (PNIEC 2021–2030) by prioritising photovoltaic generation and reducing diesel dependence. Compliance with national environmental standards and Denomination of Origin (DO) regulations ensures compatibility with wine sector quality governance. These commitments are further reinforced by EU organic certification and sustainable forest management accreditation, strengthening institutional legitimacy and market acceptance.

Overall, the PESTEL analysis shows that the success and replicability of the Bosque de Matasnos model depend not only on internal efficiency gains but also on its alignment with dynamic external drivers. By embedding circular water–energy practices within evolving policy frameworks, market dynamics, and socio-technical contexts, the case demonstrates how off-grid viticulture can anticipate regulatory

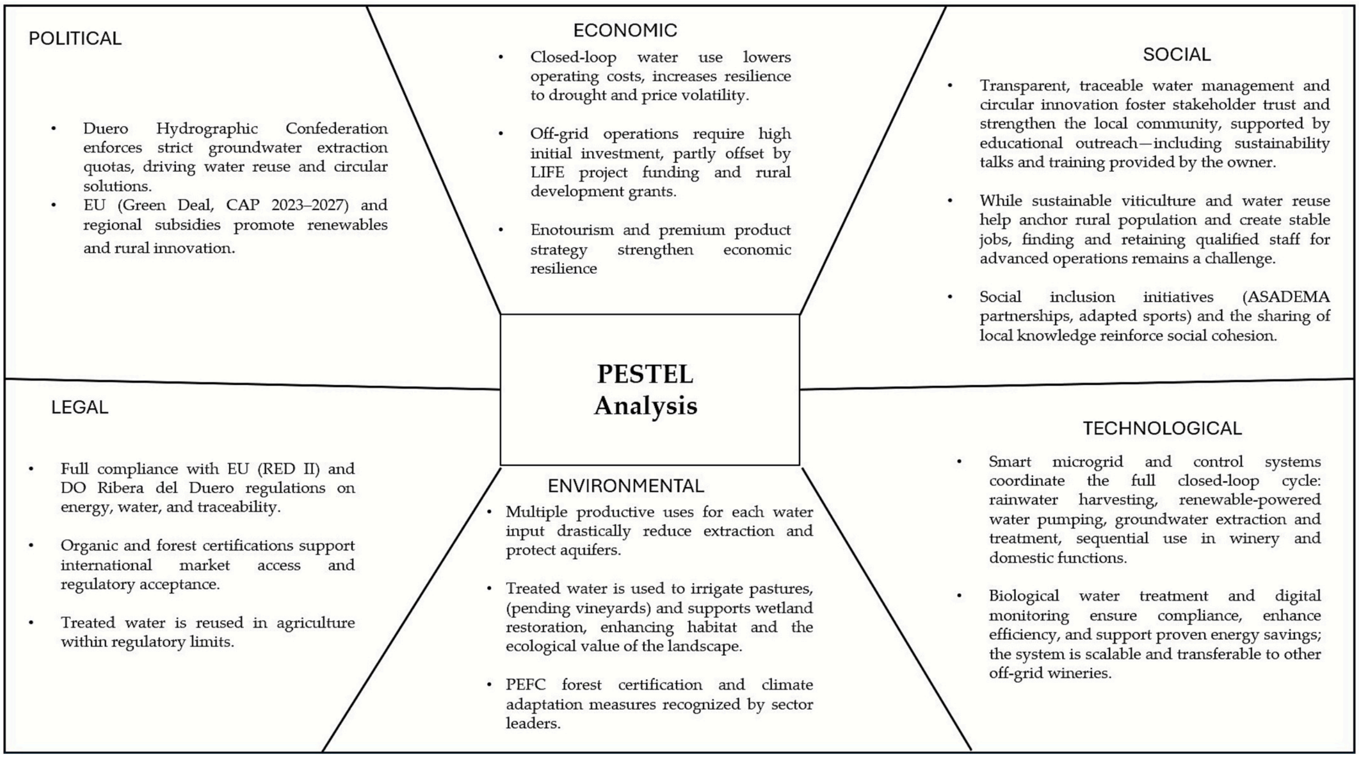

Fig. 6. PESTEL analysis of the Bosque de Matasnos circular water–energy reuse system.

tightening while maintaining competitiveness. More broadly, it provides actionable insights for policymakers, practitioners, and researchers seeking to scale circular innovations in resource-constrained agri-food systems. This confirms the value of coupling diagnostic tools such as Ecocanvas with foresight frameworks like PESTEL to guide systemic transitions (Rahman and Mishra, 2023). This multi-scalar alignment also echoes evidence from other agri-food circular initiatives, where the interplay of regulatory incentives and local socio-economic contexts is decisive for replication (Boix-Fayos and De Vente, 2023).

4. Discussion

Building on the results presented in Section 3, the Bosque de Matasnos case provides empirical evidence that circular water–energy systems can be effectively deployed in off-grid viticulture. It shows how such systems move beyond isolated measures to generate systemic value across environmental, socio-economic, and methodological dimensions. This aligns with calls to embed efficiency within broader territorial perspectives (Geissdoerfer et al., 2020).

From an environmental standpoint, Bosque de Matasnos demonstrates how circular logic can be applied through cascading water use across winery, domestic, and agricultural functions. This multifunctional loop responds to calls for aquifer sustainability as a key pillar of Mediterranean agriculture (Costa et al., 2020). Reviews highlight that treated wastewater can mitigate scarcity if monitoring safeguards soil and crop quality (Costa et al., 2025). The integration of a water-to-water heat pump powered by photovoltaics reduced energy demand and fossil fuel use, anticipating stricter carbon regulations under the European Green Deal. These results echo research on coupling emission reductions with traceability (Provenzano et al., 2024) and confirm the relevance of circularity metrics for climate adaptation (Ribeiro et al., 2025).

Compared with existing cases, Bosque de Matasnos extends circularity in viticulture. Trials in Israel and Spain confirm the potential of reclaimed water for irrigation, but also risks such as salinity accumulation (Simhayov et al., 2023). Other studies address wastewater reuse only for irrigation (Petousi et al., 2019) or consider photovoltaics and heat pumps separately (Chen et al., 2023). Much of the circular economy literature still emphasizes waste valorization (Provenzano et al., 2024). In contrast, Bosque de Matasnos integrates water reuse, renewable energy, and biodiversity stewardship in a single off-grid model, highlighting an innovation pathway not covered in previous studies and showing the value of systemic frameworks such as Ecocanvas and PESTEL.

The socio-economic dimension reinforces the role of circular models in rural resilience. Previous studies show how circular strategies generate jobs and activate value chains (Sehnem et al., 2020). Bosque de Matasnos adds nuance by linking circular practices with demographic stabilization through training, wine tourism, and partnerships with local suppliers and social enterprises (Martínez-Carrasco Pleite and Colino Sueiras, 2024). This linkage between circular practices and demographic stabilization remains underexplored in the literature, highlighting the distinctive socio-territorial leverage of viticulture.

Circular water–energy systems also create multidimensional value across markets, communities, and supply chains. For consumers, verifiable standards strengthen transparency and differentiation. For local communities, reduced aquifer pressure and drought resilience bring tangible benefits. At the organizational level, workers acquire skills in integrated resource management, and suppliers align with ESG requirements. Collaboration under LIFE Climawin supports replicability and knowledge transfer (Poponi et al., 2023).

Methodologically, the study advances Ecocanvas as a tool to integrate technical, social, and organizational dimensions into business model analysis. Its combination with PESTEL addresses the lack of multidimensional indicators and contextual integration in agri-food enterprises (Das et al., 2022). This dual approach moves beyond technical assessments to connect enterprise innovation with policy and market dynamics (Krasnokutska and Danko, 2024). To our knowledge, this is the first use of Ecocanvas and PESTEL to analyze circular water–energy integration in viticulture.

Limitations include irregular rainfall in Ribera del Duero, site-specific hydrogeology, and the need for diesel backup during peak demand. These local constraints limit generalization, but several elements are transferable, such as cascading water use, photovoltaic integration with smart control, and the use of Ecocanvas and PESTEL as systemic tools. Future research should test these components at larger scales and under different climatic conditions, including hybrid renewable configurations and long-term groundwater monitoring. Distinguishing local from replicable factors reinforces the value of this case as both a demonstration project and a reference for broader circular transitions.

Beyond site-specific outcomes, the findings carry policy relevance. By aligning enterprise innovations with the European Green Deal, the CAP (2023–2027), and EU directives on energy and water, Bosque de Matasnos shows how SMEs can anticipate regulatory tightening while enhancing competitiveness. However, this case also demonstrates that regulatory alignment does not necessarily translate into implementation readiness. Despite meeting the technical and sanitary standards required for the safe reuse of treated water, the deployment of circular water practices remains conditioned by lengthy authorisation procedures from the River Basin Authority. This reflects a broader regulatory–operational gap already noted in European assessments of water reuse implementation (Christou et al., 2024), where administrative delays hinder circular innovation despite compliance with environmental legislation.

Taken together, these results illustrate that circular transitions depend not only on technical design but also on institutional operability and policy alignment. They also resonate with broader debates on energy efficiency in winemaking (de Castro et al., 2024) and the systemic importance of water management in wineries (Matos et al., 2024). They also connect to emerging evidence on how climate extremes threaten viticulture, from increased precipitation intensity in Southern Europe (Di Carlo et al., 2019) to rapidly changing growing conditions in other wine regions such as Canada (Rayne and Forest, 2016). This echoes evidence that regulatory incentives and socio-economic contexts are decisive for replication (Boix-Fayos and De Vente, 2023). The case thus illustrates how circular water–energy practices can function not only as compliance, but also as a lever for adaptation, rural regeneration, and replicable circular innovation across agri-food systems.

5. Conclusions

This study demonstrates that circular water–energy integration in off-grid viticulture can go beyond incremental efficiency gains to operate as a systemic innovation with territorial significance. The Bosque de Matasnos case confirms that sequential water reuse, coupled with renewable-powered thermal systems and smart energy management, enhances operational autonomy while reducing diesel dependence and enabling measurable reductions in emissions. These results show that circular configurations can provide resilience in Mediterranean rural contexts facing water stress, energy vulnerability and demographic decline.

A key conclusion is that circular business models in agriculture cannot be evaluated solely on technical performance. Their viability depends equally on organizational capabilities, stakeholder alignment, territorial context and regulatory conditions. Methodologically, this study offers a replicable framework by combining Ecocanvas to structure circular value creation at enterprise level with PESTEL to interpret institutional feasibility and scalability. This multilevel approach captures interactions between technical design, business strategy and policy environment—dimensions that are often addressed separately in the literature.

From a practical perspective, the findings indicate that circular water–energy systems can be implemented with manageable investment levels, especially when leveraging existing infrastructure and phased implementation strategies. Smart energy control plays a strategic role in these configurations by optimizing renewable use, coordinating storage and load shifting, and reducing both operational costs and environmental externalities. This indicates that the transition to circularity in agri-food systems is not only a matter of technology adoption, but also of system design and decision intelligence.

From a policy perspective, the results highlight the need for regulatory frameworks that actively support circular resource integration in rural economies. European and national water regulations should operationalize safe water reuse by simplifying authorisation procedures for decentralized treatment systems, in line with EU Regulation 2020/741. Energy policy should also recognise off-grid hybrid microgrids as strategic rural infrastructure, facilitating renewable self-sufficiency through incentives for intelligent energy control and storage. In addition, rural development policies should accelerate circular business models through targeted funding mechanisms, tax incentives linked to environmental performance and dedicated programmes that support innovation in SMEs. Cross-sector policy coordination is essential, as current water, energy and agricultural regulations remain fragmented and hinder the implementation of integrated circular solutions.

At a broader level, this study aligns with major sustainability agendas. It contributes to the objectives of the European Green Deal and the EU Circular Economy Action Plan, which call for renewable integration, resource circularity and climate resilience in production systems. It also supports multiple United Nations Sustainable Development Goals (SDGs), particularly SDG 6 (Clean Water and Sanitation), SDG 7 (Affordable and Clean Energy), SDG 12 (Responsible Consumption and Production) and SDG 13 (Climate Action). By linking empirical evidence with strategic policy priorities, this work demonstrates that circular water–energy integration is not only technically feasible but also socio-institutionally viable.

Future research should quantify system-level impacts when such models are replicated at regional scale, including effects on aquifer recovery, energy resilience and territorial value creation. Further work is also needed to develop harmonised circularity indicators, business model innovation metrics and long-term monitoring protocols to support transferability across agri-food sectors. A limitation of this study is that it is based on a single case, and therefore contextual conditions may influence replicability. However, the proposed methodological approach is transferable and provides a structured basis for comparative studies in other agri-food contexts. Overall, this study shows that circular integration of water and energy in viticulture can act as a strategic lever for resource sovereignty, rural regeneration and climate adaptation in Mediterranean territories. Ultimately, the path to circular agriculture will depend not only on technological innovation, but also on the capacity to integrate resource intelligence, business model redesign and place-based governance.

CRediT authorship contribution statement

Alexy Apolo-Romero: Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Nieves García-Casarejos:** Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization. **Pilar Gargallo:** Writing – review & editing, Writing – original draft, Supervision, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the writing process of this work the authors used ChatGPT-4o in order to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Declaration of competing interest

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Data availability

Data will be made available on request.

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